

SELF-EVOLUTION SHOULD NOT DEPEND ON HOW SKILLS ARE SPLIT: AN EXACT CERTIFICATE FOR SKILL-TAXONOMY REPRESENTATION INVARIANCE

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ABSTRACT

Self-evolving language-model agents increasingly treat skill identities as control objects: a sampled skill can determine the next training task, a fresh ID can receive retrieval priority, and feedback can be written back to the selected package. We ask whether such control should change when semantic capabilities are merely split or cloned across package identities. We formulate *Skill-Taxonomy Representation Invariance* (STRI). For exact refinements, we prove that invariance is equivalent to factoring aggregate semantic-class mass through the quotient; any strictly positive identity-local score followed by normalization is therefore clone-sensitive. Released Skill-SP instantiates this failure mode at initialization: 15 packages receive probability $1/15$ each, double-supported Level-1 rows receive twice the additive eligibility opportunity of singleton rows, and a counterfactual same-state sampler-input clone changes its prompt-mixture by $TV = 0.0583$; quotient-conserved mass restores it exactly. For partial overlap, define $R^*(A; q) = \min\{t : q \leq Aw \leq tq, w, t \geq 0\}$ for support matrix A and representation-independent target $q > 0$. It equals one iff q lies in the package support cone. On Skill-SP Level-1, neutral $R^* = 2$ on full, calibration, and tool-disjoint heldout matrices. It survives every one-cell and every valid tool-block correlated stress test, but coherent package-wide support overstatement can erase the residual, making support truth an explicit load-bearing assumption; in the recorded post-rebase CPU profile, the unchanged reference LP solves the largest $16,384 \times 96$ synthetic matrix in at most 0.765 s over three runs; wall time is explicitly treated as host-load dependent. Level-3 and the 127/128-overlap logical domain are equalizable. In one archived AutoSkill P19 substrate, splitting one skill into four exact-semantic IDs deterministically crowds out a post-checkout skill and removes a mechanically defined destructive behavior (6/6 original versus 0/6 split), while ID-placebo and quotient controls restore it (3/3 each; Fisher $p = 0.00108$). Under the split representation, adding that post-checkout skill back restores the behavior in 3/3 fresh runs while a matched cleanup add-back yields 0/3 (exact one-sided Fisher = $1/20$), isolating the mediator within this substrate. Thus representation can alter executed behavior on this bounded substrate without semantic change, whereas overlap prevalence alone does not determine the cone residual. A semantic-first construction marks the action-basis escape when package-only control cannot realize q ; it is constructive, not a validated repair.

1 INTRODUCTION

Many self-evolving agents adapt through external state—memories, skills, workflows, tools, prompts, retrieval policies, or harness code—rather than only through foundation-model weights. Skills are especially attractive because a skill can package reusable procedure, metadata, examples, applicability information, and verification logic. Recent systems make the skill identity itself a control variable: Skill-SP (Huang et al., 2026) samples a skill before generating a training task, while SkillIRL (Xia et al., 2026) gives dynamically created skill IDs priority in a fixed retrieval budget.

This architecture makes the *representation* of the skill library consequential. Consider two taxonomies that expose the same executable semantic capability. One stores a primitive once; another stores an exact clone under a fresh ID. One represents generic and specific capabilities separately; another groups unchanged primitives behind a macro package. If no new primitive behavior or support region is added, should the future curriculum, retrieval exposure, or credit allocation change solely because package bookkeeping changed?

Controlled studies already vary skill availability/granularity, isolate representation/retrieval effects, decompose and recombine skills, and formalize exposure allocation (Xu & Wu, 2026; Jiang et al., 2026; Gao et al., 2026; Singh & Joachims, 2018; Biega et al., 2018). We therefore do not claim priority for those objects. Our missing object is support-fixed: should semantics-preserving package refactoring change an endogenous controller, and when can package-only mass realize a representation-independent target?

In self-evolution, package IDs are created, sampled, retrieved, and credited, so representation-only changes can alter future prompts or procedures. The challenge is to separate this bookkeeping channel from semantic scope: overlap and skill count themselves change under refactoring, requiring exact-clone quotienting, independent support truth, and a strongest same-information baseline.

We approach the question asset-first. Rather than proposing a normalization method and manufacturing a benchmark, we inspect released control surfaces, reduce the phenomenon against strongest same-information baselines, and identify a support geometry that separates positive and negative regimes.

Released Skill-SP. On API-Bank Level-1, 314 released tasks are covered by at least one released validator and 183 have multiple memberships. The release’s name+description Jaccard duplicate filter at threshold 0.33 misses all five observed specific–generic support-overlap pairs. Minimum-cardinality whole-package pruning can reduce the active library from 10 to 7 while preserving exact support, but 71 overlapping tasks remain. The residual is therefore not merely an exact-clone artifact.

Released SkillRL. In released template-mode retrieval, dynamic general skills are included before the remaining fixed $\text{top}_k=6$ budget is filled with static skills. Adding one exact-content fresh-ID copy at a time admits all 12 tested clones. Across 223 released ALFWorld task descriptions, 11/12 clone targets change the unique semantic retrieval set and 5/12 reduce the number of unique general-skill contents returned.

These systems establish representation sensitivity, but not its boundary. The logical domain provides a direct counterexample to overlap-as-sufficient-explanation: 127/128 tasks overlap while the neutral target remains exactly realizable. Thus overlap count alone cannot determine the STRI residual.

Contributions.

1. **A support-fixed systems invariant and audit object.** Building on prior studies of skill representation, routing, decomposition, and exposure, we ask the stricter question of whether a skill-indexed self-evolution controller is invariant to package refactorings that leave semantic support fixed. The novelty claim is this systems invariant and the corresponding audit object, not skill representation or exposure allocation as topics.
2. **An exact characterization of that audit object.** For package-first additive exposure, quotient factorization characterizes exact-refinement invariance and $R^*(A; q)$ decides target realizability under frozen support; a semantic-first construction marks the action-basis escape when package-only control is insufficient. These results deliberately reuse classical convex-feasibility/duality machinery: we claim the mapping from the systems invariant to an exact decidable certificate, not a new LP, cone theorem, or fairness objective.
3. **Released-system and bounded behavioral evidence with quantified boundaries.** Skill-SP and SkillRL expose identity sensitivity under released control surfaces, while positive and negative support geometries separate overlap prevalence from exact realizability. On one frozen AutoSkill P19 substrate, split/placebo/quotient and matched mediator add-back controls trace a representation-to-retrieval-to-executed-behavior consequence without promoting it to a general utility or safety theorem. We additionally quantify structured support-label misspecification and profile the unchanged reference LP up to a $16,384 \times 96$ synthetic support matrix.

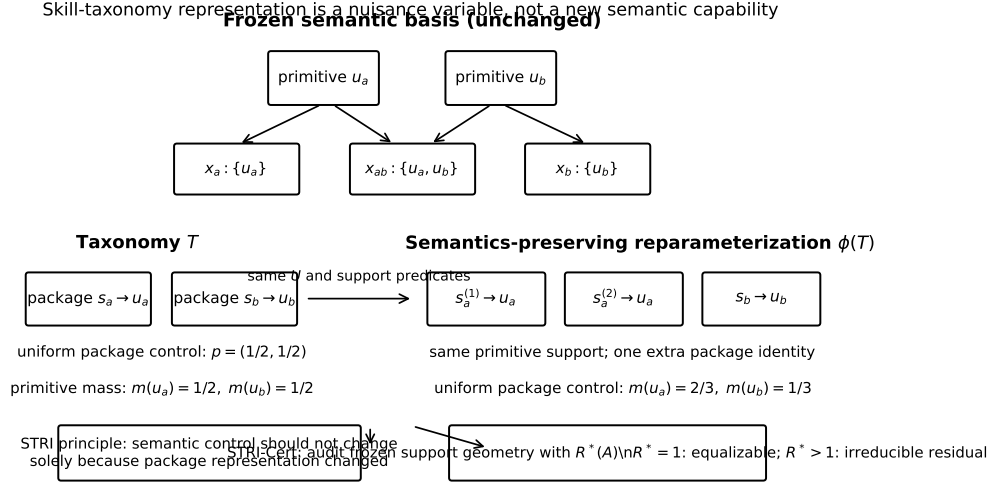


Figure 1: STRI separates semantic capability from package representation. The conceptual identity-split example leaves the primitive basis and support predicates unchanged, yet a package-indexed controller can change primitive mass solely because one primitive has two package identities. Exact clones motivate the nuisance channel; the decisive released residual in Sec. 6 survives clone reasoning and is characterized by non-equalizable partial-overlap support geometry.

2 RELATION TO EXISTING WORK

Skill systems and closest work. Prior work already studies skill granularity, representation/retrieval, composition, redundancy, safety, and co-evolution (Xu & Wu, 2026; Jiang et al., 2026; Gao et al., 2026; Zhu et al., 2026; Liu et al., 2026; Mao et al., 2026; Dong et al., 2026; Chen et al., 2026; Huang et al., 2026; Xia et al., 2026; Wang et al., 2026). STRI asks the narrower support-fixed question after exact representation nuisance is quotiented out. Skill-SP (Huang et al., 2026) is the closest support-bearing system; SkillRL (Xia et al., 2026) supplies an identity-sensitive retrieval phenotype but no independent support contract; AutoSkill (Yang et al., 2026) supplies one bounded behavioral seam, not prevalence evidence.

Optimization boundary. Fair-exposure methods already allocate attention (Singh & Joachims, 2018; Biega et al., 2018), and Eq. (1) uses classical covering/cone/duality machinery (Lovász, 1975; Sinkhorn & Knopp, 1967; Radunović & Le Boudec, 2007; Beck & Fiala, 1981; Boyd & Vandenberghe, 2004). We claim the systems invariant and audit mapping, not LP or cone novelty.

3 METHOD: FROM PACKAGE IDENTITY TO SEMANTIC EXPOSURE

3.1 PROTOCOL AT A GLANCE

STRI-Cert takes a package representation, independently frozen support A , representation-independent target q , and the controller’s pre-task information boundary, and returns an invariant/realizable/residual/fail-closed diagnosis plus a witness when available. The flow is **freeze semantics** (including uncovered-row accounting) $\rightarrow$ **quotient exact clones** $\rightarrow$ **give arbitrary non-negative package mass the full same-information support matrix** $\rightarrow$ **solve primal and dual** ($R^* = 1$ realizable; $R^* > 1$ residual) $\rightarrow$ **separate diagnosis from repair and utility**. The last step is essential: semantic-first selection is an action-basis witness, not a validated downstream controller.

3.2 FROZEN SUPPORT MATRIX

Let $X = \{x_1, \dots, x_n\}$ be a finite set of semantic contexts or tasks with independently defined first-party support truth, and let $S = \{s_1, \dots, s_m\}$ be the current skill packages. Define the binary

incidence matrix

$$A_{ij} = \mathbf{1}\{s_j \text{ supports } x_i\}.$$

The support definition is frozen before evaluating STRI. In the tool-call study, support comes from released Skill-SP validators. In the logical-reasoning study, tasks are generated by released compiler specifications, checked by the author’s fixed CSP/contract code, and cross-skill support is determined by the released compiler-alignment predicate. All certificate statements are restricted to *covered rows*, i.e., rows with at least one supported package. An uncovered row has a zero incidence row and cannot satisfy a strictly positive target constraint $q_i \leq (Aw)_i$ for any package mass; the implementation therefore excludes it from the covered-domain LP and reports coverage separately instead of silently treating infeasibility from missing support as representation distortion.

We study package-only pre-context control. At a frozen state, the controller chooses nonnegative package mass $w \in \mathbb{R}_{\geq 0}^m$ before the semantic task to be generated exists. The additive eligibility exposure of context x_i is $e_i = (Aw)_i$. This is a control-opportunity quantity, not a task probability and not utility. Let $q \in \mathbb{R}_{> 0}^n$ denote a *representation-independent semantic target*: it may encode an externally justified nonuniform priority, but it may not be changed merely because packages are renamed, cloned, split, or regrouped. When no such pre-context semantic preference is specified, we use the neutral target $q = \mathbf{1}$.

3.3 SAME-INFORMATION PRE-TASK CONTROLLERS REDUCE TO PACKAGE MASS

In released Skill-SP, skill selection occurs before task generation. At this decision point the action is one current package identity. For any fixed pre-task state z , every randomized same-information rule—including a bandit, MLP, or neural router—induces a categorical distribution over packages, hence one nonnegative mass vector $w(z)$. The strongest same-information baseline is therefore arbitrary nonnegative package mass with access to the complete frozen support matrix.

This reduction is pointwise in z . It does *not* claim that a longitudinal adaptive controller is globally equivalent to one static vector, and it does not allow a baseline to condition on the future generated task when the reviewed controller acts before that task exists.

3.4 EXACT-REFINEMENT STRI FACTORS THROUGH A QUOTIENT

Let $\mathcal{R}$ be a package representation and let $Q(\mathcal{R})$ collapse exact semantic clones—packages with identical semantics, support, and controller-relevant state—into classes. Let $G_{\mathcal{R}}$ sum a package-mass vector over these classes. For a fixed pre-context state z , define *exact-refinement STRI* by

$$G_{\mathcal{R}}p_{\mathcal{R}}(z) = G_{\mathcal{R}'}p_{\mathcal{R}'}(z) \quad \text{whenever } Q(\mathcal{R}) = Q(\mathcal{R}').$$

Proposition 1 (quotient-factorization characterization). A controller family satisfies exact-refinement STRI iff there exists a quotient-level map $\bar{p}$ such that $G_{\mathcal{R}}p_{\mathcal{R}}(z) = \bar{p}(Q(\mathcal{R}), z)$ for every representation and state.

Proof. The forward direction defines $\bar{p}$ using any representative of a quotient class; STRI makes this definition representative-independent. The reverse direction is immediate because equal quotients give equal $\bar{p}$. $\square$

Corollary 1 (identity-local normalization is clone-sensitive). Suppose $p_j = g(h_j, z) / \sum_k g(h_k, z)$ with $g > 0$ and an exact clone receives the same local state h_j . Writing its score as $a > 0$ and all other scores as $B > 0$, the semantic-class mass changes from $a/(a + B)$ to $2a/(2a + B)$, an increase of $aB/((a + B)(2a + B)) > 0$. Repeated clones drive class mass toward one, so exact-refinement invariance requires class-aware conservation or another quotient-factorized rule.

Proposition 2 (semantic-first construction handles arbitrary overlap). Normalize a target ray $q > 0$ to $\pi = q/(1^\top q)$. If every target-positive semantic unit i has at least one supporting package, choose any row-stochastic kernel K with $K_{ij} = 0$ whenever $A_{ij} = 0$, then sample $i \sim \pi$ before sampling implementation $j \sim K_i$. The semantic marginal is exactly π for arbitrary overlap, and any clone/split can redistribute K_i within the corresponding implementation class without changing that marginal.

Proof. The joint mass is $J_{ij} = \pi_i K_{ij}$, so $\sum_j J_{ij} = \pi_i \sum_j K_{ij} = \pi_i$; support-constrained row-stochastic K exists exactly when every positive-target row is covered. $\square$ This changes the action interface: it requires representation-independent semantic intent and conditional implementation. It is a constructive action-basis witness for escaping package-first infeasibility, not a same-information baseline or validated downstream method.

4 TARGET REALIZABILITY AND STRI-CERT

4.1 TARGET-CONDITIONED FINITE-SUPPORT CRITERION

Define the multiplicative distortion from the target ray q to the package support cone by

$$R^*(A; q) = \min_{w \geq 0, t \geq 0} t \quad \text{s.t.} \quad q \leq Aw \leq tq. \quad (1)$$

We write $R^*(A) \equiv R^*(A; \mathbf{1})$ for the neutral audit used in our released-system tables.

Theorem 1 (target realizability in the support cone). For any $q > 0$, $R^*(A; q) = 1$ if and only if $q \in \text{cone}(A) = \{Aw : w \geq 0\}$. Otherwise $R^*(A; q) > 1$ is the tight minimum componentwise multiplicative distortion from q achievable by package-only mass.

Proof. The constraints force $t \geq 1$. At $t = 1$ they reduce exactly to $Aw = q$, which is equivalent to $q \in \text{cone}(A)$. For any feasible w , the smallest admissible t is $\max_i (Aw)_i / q_i$ after scaling w so that $\min_i (Aw)_i / q_i = 1$; minimizing over w gives the tight distortion. $\square$

This separates STRI from a uniformity assumption: $q = \mathbf{1}$ is the neutral target, not the definition of representation invariance. It also makes exact clone/split invariance immediate: duplicating a support column does not change $\text{cone}(A)$, so it cannot change $R^*(A; q)$ for any fixed q . The cone-membership statement itself is classical convex feasibility; STRI contributes the system invariant that makes this feasibility question the relevant audit object for a self-evolving skill controller, not a new theorem about convex cones.

Proposition 3 (dual certificate). The dual of Eq. (1) is

$$D^*(A; q) = \max_{\alpha, \beta \geq 0} q^\top \alpha \quad \text{s.t.} \quad A^\top (\beta - \alpha) \geq 0, \quad q^\top \beta \leq 1. \quad (2)$$

Strong duality gives $D^*(A; q) = R^*(A; q)$. Thus every dual-feasible pair is a certified lower bound, while an optimum is complete even when no local witness template is available. On every audited neutral-target regime the numerical primal–dual gap is below 10^{-8} .

4.2 HUMAN-READABLE FACTOR-2 WITNESS

Theorem 2 (global-singleton overlap lower bound). Suppose packages a, b each have a context supported by exactly that one package over the full frozen support universe, and another context is supported by both a and b , possibly with additional packages. If both singleton contexts retain positive exposure, then every nonnegative package mass has max/min exposure ratio at least two on the focal rows, and therefore $R^*(A) \geq 2$ for the full support matrix.

Proof. Let $r > 0$ be the minimum exposure among the two singleton contexts and the shared context. Since the first singleton support signature is exactly $\{a\}$, its exposure is $w_a \geq r$; likewise $w_b \geq r$. The shared context contains both a and b , and additional packages have nonnegative weight, so its exposure is at least $w_a + w_b \geq 2r$. Thus every w has focal max/min ratio at least two. For the same w , the full-matrix maximum is no smaller than the focal maximum and the full-matrix minimum is no larger than the focal minimum, so the full-matrix ratio is also at least two. Minimizing over w gives $R^*(A) \geq 2$. $\square$

For the neutral target $q = \mathbf{1}$, Theorem 2 is also a sparse dual certificate: put $\alpha = 1$ on the two singleton rows, $\beta = 1$ on their shared row, and zero elsewhere. Dual feasibility follows from the support pattern and the objective is two. In released Skill-SP Level-1, the LP dual recovers such a sparse optimum and the primal attains $R^* = 2$.

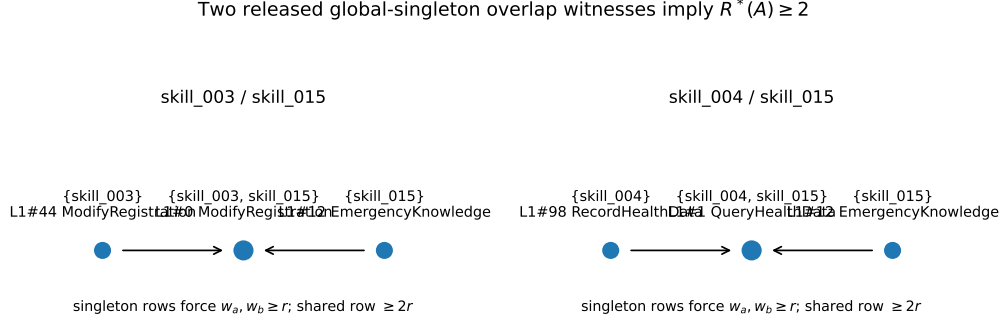


Figure 2: Two released global-singleton overlap witnesses. In each case, retaining positive exposure on the two singleton regions forces at least twice that minimum exposure on the shared region.

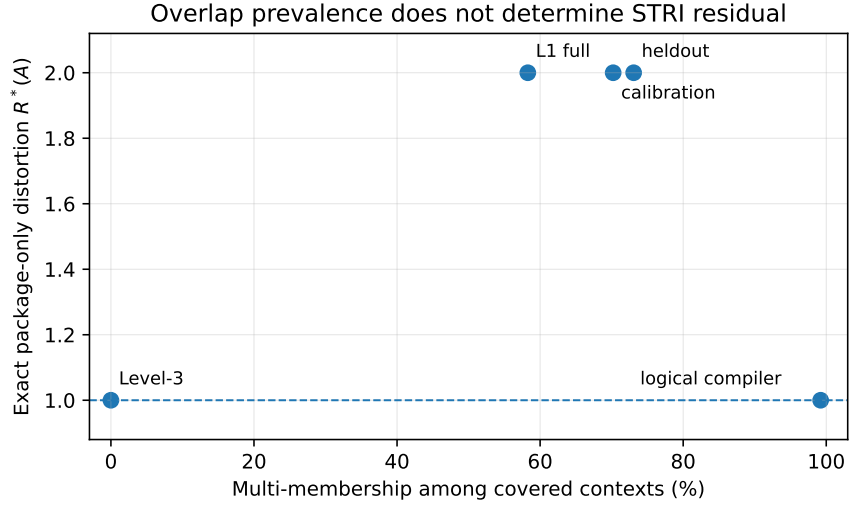


Figure 3: Overlap prevalence does not determine STRI residual. The three Level-1 regimes independently have $R^*(A) = 2$, while Level-3 is equalizable with disjoint support and the logical compiler domain is equalizable despite 127/128 multi-membership tasks.

4.3 STRI-CERT AND DEPLOYABLE VARIANTS

STRI-Cert is a diagnostic protocol, not a learned predictor or a new LP solver: freeze independently defined support and target q ; solve Eq. (1) and its dual; report target-realizable iff $R^* = 1$, otherwise the tight residual; expose a sparse witness when available; and fail closed when support, target provenance, or optimization is invalid. The reference implementation uses SciPy’s HiGHS backend for both primal and dual programs; paper-level classification uses tolerance 10^{-8} , and all audited primal–dual gaps are at most 10^{-8} . Degenerate optima do not affect the certificate because the reported object is the optimal value and induced exposure feasibility, not a unique package-weight vector. Arbitrary nonnegative global package reweighting already receives the complete frozen support matrix, making text deduplication, clone removal, uniform routing, and scalar retuning weaker same-information reductions.

Two extensions make further assumptions explicit. For graded support with independently justified $0 \leq L \leq A \leq U$, the exact box-robust audit replaces Eq. (1) by $Lw \geq q$ and $Uw \leq tq$; estimating valid bounds remains a separate identification problem. Deployment constraints can likewise be audited rather than ignored: adding $w_j \leq \rho \sum_k w_k$ for every package defines a max-share constrained $R_\rho^*(A; q) \geq R^*(A; q)$. Both remain linear programs.

Table 1: Two-stage audit across released systems. We test observable identity sensitivity first and run STRI-Cert only when independent support truth exists; absence of a support contract is not repaired with candidate-defined labels.

System	Identity perturbation	Independent A ?	Audit role
Skill-SP	Pristine sampler gives $2\times$ eligibility; sampler-input clone changes class mass $1/15 \rightarrow 1/8$, while quotient conservation restores it exactly	Yes: released validators	Controller violation + exact clone repair + geometry certificate
SkillRL	Fresh-ID exact clones: 11/12 change semantic retrieval over 223 tasks	No explicit support contract	Phenotype + qualified downstream bridge; no R^*

Table 2: Matched baseline ladder on the Skill-SP tool-call support matrix. Every baseline uses the same frozen released support information; the ladder deliberately strengthens from name-level duplicate removal to the exact best context-independent package reweighting.

Baseline / reduction	Outcome	Evidence
Released name+description Jac-card duplicate filter	Does not absorb	Misses all 5/5 observed specific-generic support-overlap pairs; similarity 0.1429–0.20 is below the released 0.33 threshold.
Exact semantic-clone deduplication	Does not absorb partial overlap	The positive regime uses distinct specific/generic packages with different unique support regions.
Minimum-cardinality whole-package pruning preserving exact support	Partial reduction only	Active packages drop from 10 to 7, but 71 overlap rows remain while exact support is preserved.
Arbitrary nonnegative global package weights	Exact neutral-target residual	The exact optimum is $R^*(A; \mathbf{1}) = 2$.
Global-singleton overlap witness	Sparse dual lower bound	Two independent released witnesses imply $R^*(A; \mathbf{1}) \geq 2$, and the exact LP attains 2.

5 EXPERIMENTAL SETUP

Evidence, estimands, and controls. Skill-SP Level-1 is the primary positive support regime; a frozen tool-disjoint split tests transfer and Level-3 is a disjoint negative control. A first-party logical compiler supplies a high-overlap programmatic negative regime. SkillRL is restricted to its released retrieval seam because it lacks independent support labels; AutoSkill contributes one frozen P19 representation intervention with mechanical behavior and matched placebo/quotient/add-back controls. The structural estimand is neutral $R^*(A; \mathbf{1})$ with primal/dual tolerance 10^{-8} , and the strongest same-information baseline is arbitrary nonnegative reweighting with the complete frozen A .

Six evidence roles. E1 is the released-system main result; E2 quotient/baseline ablations; E3 tool-disjoint, negative-regime, and bounded cross-system evidence; E4 diagnostic/support sensitivity; E5 sparse witnesses and the P19 mediator trace; E6 conditional solver cost. The added offline completion step uses E4/E6 analyses on frozen data or synthetic matrices—zero new support annotations, task outcomes, model calls, or GPU runs.

6 EXPERIMENTS AND RESULTS

6.1 SKILL-SP TOOL-CALL SUPPORT AND RELEASED-CONTROLLER CONSEQUENCE

API-Bank Level-1 has 314 covered rows, 183 with multiple memberships, and neutral $R^* = 2$. The released pre-task sampler initializes all 15 tool-call packages at probability $1/15$, so a double-supported row receives $2/15$ eligibility versus $1/15$ for a singleton. The source paper reports a -1.9 point accuracy change when dynamic routing is replaced by uniform routing (Huang et al., 2026); we use this only to establish that allocation is operational, not to attribute utility to STRI.

Table 3: Neutral-target ($q = 1$) mechanism boundary across real first-party support regimes. High overlap is neither necessary nor sufficient; the discriminator is target realizability in the support cone.

Regime	Covered	Multi	Overlap	R^*	Certificate
API-Bank Level-1 full	314	183	58.3%	2.0	Residual
Level-1 calibration tools	47	33	70.2%	2.0	Residual
Level-1 tool-disjoint heldout	52	38	73.1%	2.0	Residual
API-Bank Level-3	34	0	0.0%	1.0	Equalizable
Logical compiler validation	128	127	99.2%	1.0	Equalizable

For a same-state sampler-input reparameterization, we copy one active package with identical content, support, statistics, and sampler state under a fresh ID and rerun the author’s weighting function. This is not a claim that the induction path would admit a literal duplicate. The cloned semantic class moves from $1/15$ to $2/16 = 1/8$, yielding prompt-mixture $TV = 7/120 = 0.0583$ because package ID is absent from the questioner message. Quotient-conserved mass instead splits the original $1/15$ class mass and restores prompt TV exactly to zero. These are input-mixture statements, not generated-task or utility claims.

A frozen tool-disjoint test gives the same structural answer: calibration has 47 covered rows/33 multi-memberships and $R^* = 2$; heldout has 52/38 and $R^* = 2$. The two released factor-2 witnesses are `skill_003/skill_015` (singleton L1#44, L1#12; shared L1#0) and `skill_004/skill_015` (singleton L1#98, L1#12; shared L1#1), matching the exact LP.

6.2 LEVEL-3: DISJOINT-SUPPORT NEGATIVE CONTROL

Level-3 provides a natural released negative regime. Thirty-four rows are covered, every covered row has exactly one skill membership, and $R^* = 1$. Equal weights on the active packages equalize every covered task. STRI-Cert therefore does not mechanically label every released taxonomy problematic.

6.3 LOGICAL COMPILER DOMAIN: HIGH OVERLAP BUT NO RESIDUAL

The first-party logical release gives a stronger negative control: 128/128 programmatically generated units pass the author’s compiler/uniqueness contracts and are cross-checked against all eight released support specifications, with no language model. Although 127/128 units have multiple memberships across 38 support patterns, $R^* = 1$ because `skill_008` supports every unit. This equalizer is concentrated: the max-share audit gives $R_{0.75}^* = 4/3$, $R_{0.5}^* = 2$, and $R_{0.25}^* = 4$. High overlap is therefore not sufficient; realizability depends on support geometry and the declared feasible package-mass set.

6.4 SKILLRL: INDEPENDENT PHENOTYPE, BOUNDED ROLE

SkillRL confirms identity sensitivity in a different control surface: fresh-ID exact-content clones are admitted for 12/12 tested skills, 11/12 change semantic retrieval over 223 ALFWorld descriptions, and 5/12 reduce unique general-skill content under `top_k=6`. It does not expose independent support truth for R^* , so we do not retrofit a candidate-defined A ; a qualified endpoint bridge found no success disagreement beyond placebo and is not a population-level no-effect theorem.

6.5 AUTO-SKILL: DYNAMIC BEHAVIORAL PROPAGATION

AutoSkill (Yang et al., 2026) exposes a first-party `sdk.search` seam. In one frozen five-skill P19 library at `top_k=5`, a four-way exact-semantic split crowds out a post-checkout skill; ID renaming preserves, and exact-semantic quotienting restores, the original semantic set. The mechanical post-checkout behavior occurs in 6/6 original, 0/6 split, 3/3 placebo, and 3/3 quotient runs (one-sided Fisher $p = 0.00108$). Under the split, adding the crowded-out skill back restores behavior in 3/3 fresh runs versus 0/3 for a matched cleanup add-back ($p = 1/20$). This isolates one mediator on one substrate, not AutoSkill utility, safety, or prevalence.

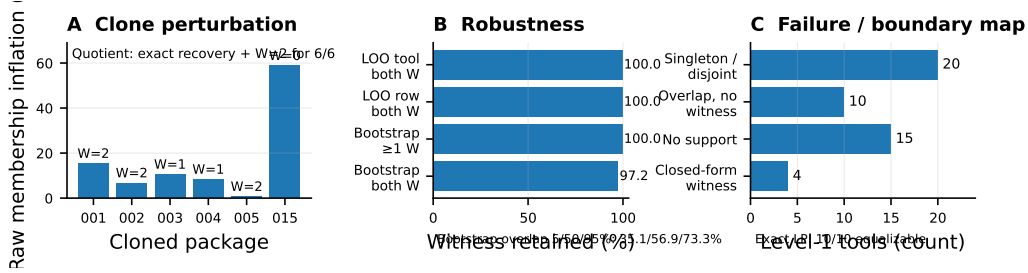


Figure 4: Representation-ablation and robustness evidence. **A**: clone-induced raw multiplicity/witness changes vanish under exact-support quotienting. **B**: factor-2 witnesses survive deletion tests and 500 tool resamples. **C**: the simple witness is only a template; exact LP subsequently resolves the ten overlap/no-witness tools as equalizable.

6.6 MATCHED BASELINES, REPRESENTATION ABLATIONS, AND FAILURE ANALYSIS

Table 2 gives the same-information ladder. Minimum exact-coverage pruning removes 61.2% of Level-1 overlap (183 to 71 multi-membership rows), yet arbitrary nonnegative package mass with the full support matrix still has $R^* = 2$ on full, calibration, and heldout regimes. Exact column cloning/splitting leaves the support cone and $R^*(A; q)$ unchanged; quotienting restores the original support rows and interpretable witness layer (Figure 4A).

Across 49 Level-1 tools, 4 have the closed-form witness and all are residual; all 10 overlap-without-witness tools are resolved equalizable by the exact LP (Figure 4C). One-cell stress is also asymmetric and explicit: all 1,387 additions and all 366 non-uncovering deletions keep Level-1 residual, whereas Level-3 turns residual under every one of 102 additions and the logical domain breaks under 127/595 valid deletions. Earlier leave-one-out and 500 fixed-seed tool bootstraps remain consistent.

E4: structured support misspecification. Correlated stress sharpens the support assumption. All 184 valid tool-block $\times$ package biases remain residual and 49/56 observed-signature block relabels remain residual. Package-wide bias is the failure boundary: only 3/9 coverage-preserving variants remain residual, and adding `skill_015` support to its 22 missing rows already makes the matrix equalizable. Thus local and tested tool-local errors are tolerated, but coherent package-wide overstatement need not be.

6.7 E6: CONDITIONAL SOLVER COST

On deterministic covered synthetic matrices (three seeds/size), the recorded post-rebase batch of the unchanged dense SciPy/HiGHS Eq. (1) solver grows from 0.00234 s median at 128×8 to 0.685 s at $16,384 \times 96$; the largest-case maximum is 0.765 s. That LP has 97 variables, 32,768 inequalities, and 24.25 MiB dense coefficients. Wall time is environment/load-specific; this conditions on already-materialized support and makes no support-acquisition, asymptotic, or solver-novelty claim.

7 DISCUSSION, LIMITATIONS, AND TAKEAWAY

STRI-Cert is exact conditional on A and q , not a certificate that support labels are correct; the 22-cell package-wide false-positive boundary makes this assumption concrete, and learned support would require independent calibration or justified box bounds. AutoSkill identifies one same-run P19 representation-to-behavior path, not population utility, safety, or longitudinal regret; a broader claim requires a separately preregistered post-query intervention. SkillRL remains an identity-sensitive phenotype without an independent A , and AutoSkill remains a bounded substrate rather than prevalence evidence. The CPU profile is conditional on materialized support, while semantic-first selection is a constructive action-basis witness rather than a validated repair. Practical takeaway: quotient representation nuisance, test $R^*(A; q)$, and repair only the assumption or interface the certificate identifies.

AI USE STATEMENT

In this work, generative AI tools assisted with conceptual framing and hypothesis refinement; formulation and critique of mathematical claims and proof drafts; experiment and falsifier design; implementation and review of analysis, experiment-control, and manuscript-quality-assurance code; interpretation of intermediate results; primary-source literature triage; and preparation or editing of manuscript text, figures, tables, and references. Generative models were also used as the task proposer in a preregistered Qwen3-8B dynamic pilot; that pilot failed its frozen author-native qualification gate, remains documented in the supplement, and is excluded from scientific claims about dynamic STRI. Load-bearing numerical claims in the submitted narrow paper are bound to released first-party artifacts or deterministic/programmatic analysis outputs. The support certificate, theorem edge cases, transition logic, and paper-level numerical/citation constraints were exercised by executable tests; related-work boundaries and bibliographic metadata were checked against primary or official sources. AI-assisted outputs were not treated as independent scientific authority. We take responsibility for the final content of this work, including text, claims, code, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The main paper states the exact-refinement quotient characterization, semantic-first design construction, target-conditioned primal/dual programs, closed-form witness, box-robust extension, and constrained-controller audit used for the load-bearing claims. The Skill-SP controller audit is bound to author commit `bb693c89fee66e1f824d6a777759a49b7a295a83`, released `library.py` SHA-256 `8441de35...e2756e0`, question-generation SHA-256 `1b4cb67f...82d739b`, initial package-bundle SHA-256 `d4c4350a...68b6e7`, and tool-call membership SHA-256 `b73d8637...6f2e17d4`. The supplement packages our controller-audit/certificate code, tests, frozen receipts and matrices, but intentionally does not redistribute the third-party Skill-SP repository; its README gives the exact author commit required for end-to-end rerun. LPs use SciPy HiGHS; certificate classification uses tolerance 10^{-8} . The offline completion analysis is implemented in `asset_first_stri_offline_completion_analysis_20260824.py` and records the frozen Level-1 structured-misspecification families plus deterministic synthetic R^* CPU profiling in a content-addressed JSON artifact. Executable tests cover the target-conditioned dual, target-ray scaling, semantic-first target realization under arbitrary overlap, quotient clone invariance, sampler-weight recomputation, literal-duplicate rejection, exact questioner-message equality, the 7/120 prompt-mixture shift, quotient restoration to zero, one-cell support perturbations, per-tool LP results, and the offline analysis contract. The added E4/E6 analyses use no new support annotations, task outcomes, model calls, or GPU runs. All dynamic thresholds were frozen before outcomes. The failed Qwen3-8B bank remains a transparent non-result; the qualified SkillRL bridge separately records 18/24 competence calibration, 24 paired A/B/C/D units, exact A/D coupling, and the statistical-resolution boundary that prevents upgrading a realization STOP into a population-level no-effect claim.

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